

Moon Spiral Memory Comparison

NEMo / Terrain-Nerfacto

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1 Summary

Dataset inspected: `/home/shared/data_nerfstudio/moon_spiral_2_masked`. The path `/home/shared/data_nerfstudio/moon.s` was not present during inspection.

The image data dominates loaded training memory. The main 100 PNG images occupy 385 MiB on disk, but expand to 791 MiB when decoded as uint8 RGBA, or 2.37 GiB if held as float32 RGB tensors.

The GT DEM point cloud is 4,000,000 XYZ float64 points, 91.6 MiB. As the raster it actually is, the height-only representation is much smaller: 30.5 MiB as float64, 15.3 MiB as float32, or 7.6 MiB as uint16/int16 quantized heights before compression.

The learned NEMo height field is tiny compared with the radiance field. The standalone `nemo_model.pt` is about 103 KiB on disk and contains 24,963 tensor parameters, about 0.095 MiB. A single Nerfstudio checkpoint is about 168 MiB on disk.

For a realistic mission-side terrain product, a 10 cm quantized height raster with origin, spacing, vertical offset, and scale metadata is a good default: it compressed to 1.26 MiB as tiled ZSTD GeoTIFF for this DEM, with 2.9 cm RMSE and less than 5 cm max height error relative to the float64 DEM heights.

2 Dataset Footprint

Item	Disk size	Loaded or raw memory	Notes
Full masked dataset folder	1.6 GiB	variable	Images, COLMAP copies, DEM point clouds
Main image set	385 MiB	791 MiB uint8 RGBA; 2.37 GiB float32 RGB	100 images, 1920 x 1080
COLMAP full-res images	460 MiB	791 MiB uint8 RGBA; 2.37 GiB float32 RGB	Another 100 full-res images
COLMAP <code>images_2</code>	119 MiB	198 MiB uint8 RGBA; 593 MiB float32 RGB	960 x 540
COLMAP <code>images_4</code>	28.8 MiB	49.4 MiB uint8 RGBA; 148 MiB float32 RGB	480 x 270
COLMAP <code>images_8</code>	7.0 MiB	12.4 MiB uint8 RGBA; 37.0 MiB float32 RGB	240 x 135
GT DEM point cloud	91.6 MiB	91.6 MiB	Shape (4,000,000, 3), float64
COLMAP point cloud	1.23 MiB	1.23 MiB	Shape (53,558, 3), float64

3 Model and Output Footprint

Example checkpoint: `2026-05-16_threshold-vertical-fix_5000/nerfstudio_models/step-000019999.ckpt`.

Item	Disk size	Tensor memory	Notes
Nerfstudio checkpoint file	168 MiB	68.5 MiB model + 105.8 MiB optimizer	Includes training state
Main radiance field tensors	in checkpoint	46.6 MiB	<code>_model.field.*</code>
Proposal network tensors	in checkpoint	12.4 MiB	<code>_model.proposal_networks.*</code>
Other model tensors	in checkpoint	9.43 MiB	Misc model state
Camera optimizer tensors	in checkpoint	0.002 MiB	588 parameters
Height field tensors	in checkpoint	0.095 MiB	24,963 parameters
Standalone NEMo height model	103 KiB	0.095 MiB	Same height-field payload

Repeated experiment artifacts under `terrain-nerfacto` dominate storage:

Artifact type	Count	Total size
<code>.ckpt</code> files	35	5.34 GiB
<code>nemo_model.pt</code> files	28	2.8 MiB
PNG render/plot artifacts	4,460	1.73 GiB
HTML Plotly artifacts	38	227 MiB
JSON files	49	504 KiB

4 GT DEM Raster Structure

The GT DEM point cloud is already a regular raster grid:

Property	Value
Points	4,000,000
Grid shape	2000 x 2000
X range	-1000.0 m to 1000.0 m
Y range	-1000.0 m to 1000.0 m
Grid spacing	1.000500250125 m
Z min	-42.171206 m
Z max	83.035019 m
Z mean	23.420581 m
Z standard deviation	27.015585 m

Because x and y are regular, storing all XYZ points is redundant. A raster representation only needs a 2000 x 2000 height array, origin, x/y spacing, vertical offset and scale if quantized, and coordinate frame metadata.

5 GT DEM Raster Memory

Raster representation	Raw memory	Loss relative to float64 DEM
Height-only float64 raster	30.5 MiB	None
Height-only float32 raster	15.3 MiB	RMSE 0.0000009 m, max 0.0000038 m
Height-only float16 raster	7.6 MiB	RMSE 0.00735 m, max 0.0311 m
uint16, 1 cm quantization	7.6 MiB	RMSE 0.00289 m, max 0.00498 m
uint16, 5 cm quantization	7.6 MiB	RMSE 0.0144 m, max 0.0249 m
uint16, 10 cm quantization	7.6 MiB	RMSE 0.0289 m, max 0.0498 m
uint16, 25 cm quantization	7.6 MiB	RMSE 0.0721 m, max 0.1245 m
uint16, 50 cm quantization	7.6 MiB	RMSE 0.1443 m, max 0.2490 m
uint16, 1 m quantization	7.6 MiB	RMSE 0.2887 m, max 0.4981 m

Quantized rasters use:

```
stored_value = round((height_m - z_offset_m) / z_scale_m)
height_m = stored_value * z_scale_m + z_offset_m
z_offset_m = -42.17120622568093
```

6 GT DEM Compression Results

Measured on the actual 2000 x 2000 DEM height raster.

Representation	Compressed format	Size	RMSE	Max abs error
float64 height raster	.npy + zstd -19	6.15 MiB	0 m	0 m
float32 height raster	.npy + zstd -19	5.79 MiB	0.0000009 m	0.0000038 m
float32 height raster	tiled GeoTIFF ZSTD	5.05 MiB	0.0000009 m	0.0000038 m
float16 height raster	.npy + zstd -19	3.87 MiB	0.00735 m	0.0311 m
uint16, 1 cm	tiled GeoTIFF ZSTD	3.19 MiB	0.00289 m	0.00498 m
uint16, 5 cm	tiled GeoTIFF ZSTD	1.75 MiB	0.0144 m	0.0249 m
uint16, 10 cm	tiled GeoTIFF ZSTD	1.26 MiB	0.0289 m	0.0498 m
uint16, 25 cm	tiled GeoTIFF ZSTD	0.78 MiB	0.0721 m	0.1245 m
uint16, 50 cm	tiled GeoTIFF ZSTD	0.54 MiB	0.1443 m	0.2490 m
uint16, 1 m	tiled GeoTIFF ZSTD	0.37 MiB	0.2887 m	0.4981 m

7 Mission-Side Recommendation

Setting	Recommendation
Raster shape	2000 x 2000
Horizontal spacing	1.000500250125 m
Stored type	uint16 or int16
Vertical scale	0.10 m
Vertical offset	-42.17120622568093 m
Compression	ZSTD with horizontal predictor
Expected size for this DEM	1.26 MiB
Expected error	2.9 cm RMSE, 5.0 cm max absolute error

This is a strong default because the quantization error is far below the current NEMo GT DEM error scale and below the level that would matter for the path-planning experiments described here. If the mission requires more conservative archival quality, use 1 cm or 5 cm quantization. If the terrain map is only for coarse navigation, 25 cm quantization compresses below 1 MiB while keeping max height error near 12.5 cm.

8 Practical Takeaways

- The full XYZ float64 DEM point cloud is not a storage-efficient mission format. It costs 91.6 MiB because x and y are repeated for every cell.
- A float32 height raster is already a near-lossless replacement at 15.3 MiB raw or 5.05 MiB compressed GeoTIFF.
- A 10 cm quantized raster is the best size/quality tradeoff from these measurements: 1.26 MiB compressed with centimeter-scale error.
- The NEMo height-field model is much smaller than any high-resolution raster: about 103 KiB on disk. That compactness is useful, but it is a learned approximation rather than a direct terrain product.
- Storage pressure in the current output tree comes from repeated Nerfstudio checkpoints and rendered PNGs, not from NEMo height-field weights.